

Project Overview:

This project aims to classify EEG signals from single trials using machine learning techniques to distinguish between *target* and *distractor* stimuli in normal and schizophrenia subjects. The goal is to identify meaningful differences in brain responses, especially P300 event-related potentials (ERPs), and explore whether brain connectivity features perform good over conventional ERP features in classification accuracy.

Methodology:

- **Dataset:** 108 subjects (54 normal, 54 schizophrenia) recorded via 64-channel EEG under the Auditory P300 oddball paradigm.
- **Feature Extraction:**
 - ERP Features:* Peak amplitude and latency from Pz, Cz, and Fz channels.
 - Connectivity Features:* Graph theory metrics using the Brain Connectivity Toolbox ex. degree, clustering coefficient, efficiency, etc.
- **ML Techniques:** Classification algorithms using scikit-learn.
- **Evaluation:** Accuracy, feature importance comparison across classifiers and feature types.

Tools & Technologies: MATLAB, Python, Brain Connectivity Toolbox (MATLAB), scikit-learn, pandas

Key Results:

- Classification models trained with connectivity features generally outperformed those using ERP features, especially when experimented to larger subject groups.
- The best classification accuracy was observed for the 54-subject group using connectivity features, demonstrating the power of network-based analysis.

Files included:

Complete thesis doc: EEG_Classification_Thesis_Script.pdf and Python code